EXERCISE 10.2

Find the derivatives of the following functions with respect to corresponding independent variables:

(1) $f(x) = x - 3\sin x$ (2) $y = \sin x + \cos x$

(3) $f(x) = x \sin x$ (4) $y = \cos x - 2\tan x$

(5) $g(t) = t^3 \cos t$ (6) $g(t) = 4\sec t + \tan t$

(7) $y = e^x \sin x$ (8) $y = \dfrac{\tan x}{x}$

(9) $y = \dfrac{\sin x}{1+\cos x}$ (10) $y = \dfrac{x}{\sin x + \cos x}$

(11) $y = \dfrac{\tan x - 1}{\sec x}$ (12) $y = \dfrac{\sin x}{x^2}$

(13) $y = \tan\theta(\sin\theta + \cos\theta)$ (14) $y = \operatorname{cosec} x \cdot \cot x$

(15) $y = x \sin x \cos x$ (16) $y = e^{-x} \cdot \log x$

(17) $y = (x^2 + 5)\log(1+x)e^{-3x}$ (18) $y = \sin x^\circ$

(19) $y = \log_{10} x$ (20) Draw the function $f'(x)$ if $f(x) = 2x^2 - 5x + 3$

10.4.2 Examples on Chain Rule

Example 10.8

Find $F'(x)$ if $F(x) = \sqrt{x^2+1}$.

Solution

Take $u = g(x) = x^2 + 1$ and $f(u) = \sqrt{u}$

$\therefore F(x) = (fog)(x) = f(g(x))$

Since $f'(u) = \dfrac{1}{2}u^{-\frac{1}{2}} = \dfrac{1}{2\sqrt{u}}$ and

$g'(x) = 2x$, we get

$F'(x) = f'(g(x))g'(x)$

$= \dfrac{1}{2\sqrt{x^2+1}}.2x = \dfrac{x}{\sqrt{x^2+1}}$.

Example 10.9

Differentiate : (i) $y = \sin(x^2)$ (ii) $y = \sin^2 x$

Solution

(i) The outer function is the sine function and the inner function is the squaring function.

Let $u = x^2$

That is, $y = \sin u$.

Therefore, $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$

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